Algebra 1

Part 1

Alveary

Alveary

Math

Algebra 1: Part 1

GRADE(S): 8

About the Course

Students discover and practice algebraic concepts through real-world narratives and problem-solving.

This topic is included in the following course(s): Mathematics: Grade 8

Mathematics: Grade 8
For students in elementary school, we recommend RightStart Math (Second Edition), because a) it adheres to Mason's principle that the idea always comes before the algorithm, b) it reflects the very best current research in how children learn mathematics, and c) it offers much-needed support for teachers whose own math instruction may have been lacking. Students in middle school take RightStart Geometry and also add in Pre-Algebra (Hands-On Equations) and Jacobs' Elementary Algebra.

The lessons should be done in the morning and games in the afternoon. Both mental math and games are critical parts of math instruction, so do not skip them. These lessons are paced out to help keep you on track. There are a few catch-up days built in and can be used to finish incomplete work or to review concepts. The general goal should be to complete one lesson per day. It is important that you complete one level per year, even if that means continuing during breaks, as moving slower can affect high school credits later.

Placement & Combining Tips

Even students in Grade 8 who have not taken Pre-Algebra should take this course to keep on pace for earning high school credits.

Prerequisite: RightStart F or mastery of basic arithmetic, fractions, decimals, and percents.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
8	Algebra 1: Part 1 3 times/week 40 min	Elementary Algebra Student Book Elementary Algebra Teacher Guide (Revised Edition)

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Mathematics: Grade 8				
Algebra 1: Part 1 Math Games: Grades 1-8	Geometry: Grade 8 Math Games: Grades 1-8	Algebra 1: Part 1 Math Games: Grades 1-8	Geometry: Grade 8	Algebra 1: Part 1

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

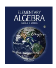
Mathematics: Grade 8
☐ If you downloaded the RightStart ebook, print out enough copies of each page and put them in a binder for easy access.

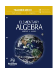
☐ Organize math manipulatives for easy access.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

Algebra 1: Part 1


Elementary Algebra Student Book


Elementary Algebra Teacher Guide (Revised Edition)

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
(No Subject Supplies Assigned)

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

∞ Foundations (See Section 10: Math)

Algebra 1: Part 1

How To Teach

Read

The RightStart Math Level G materials are written to the student and may be done independently. However, you may still find it very helpful to work through the lessons to familiarize yourself with the concepts.

Prepare

For Pre-Algebra, it is highly recommended that students be guided by a teacher during these lessons. Teachers may watch the video lessons to learn the concepts and how to teach them to students. If you choose to have your student work independently using the video lessons, you may still find it very helpful to work through the lessons yourself to familiarize yourself with the concepts. Teachers must stay apprised of how students are progressing and be ready to step in and help, since problems can compound quickly. If you are working through the lessons ahead of your student, note the places where you have to stop and think, because those are probably going to be places where students are going to be challenged, as well.

Growth Mindset

Resist the temptation to fall back on just teaching a memorizable procedure when the student struggles with a concept. Growth happens in the struggle. Work on resilience by modeling a growth mindset.

Leave & Return

When your student hits a wall during a lesson, stop and move on to the next subject in your day or take a break. Explain that even when you stop working on a problem, your brain continues to work on it while you do other things, and you don't even know it. Many times, when you come back to the problem the next day, it comes more easily.

Connect

Look for ways to bring math into everyday life and conversation "by the way" - through cooking, gardening, house projects, health, games, etc. Students should discover how math is useful and beautiful, and how it helps us explain and navigate the world.

Algebra 1: Part 1

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Term 1

WEEK 1 ⬚ **40m Algebra 1: Part 1 - Lesson 1**
A Number Trick

Materials: Jacobs Algebra

➔ READ, NARRATE, & DISCUSS
Introduction p.1-3
"Think of a " - "such as these."

➔ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Exercises p.3-4

WEEK 1 ⬚ **40m Algebra 1: Part 1 - Lesson 2**
Addition

Materials: Jacobs Algebra

➔ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 1 p.6-8
"Soon after" - "any further."

➔ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.8-9

WEEK 1 ⬚ **40m Algebra 1: Part 1 - Lesson 3**
Subtraction

Materials: Jacobs Algebra

➔ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 2 p.11-12
"The month of" - "from the first."

➔ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.12-14

WEEK 2 ⬚ **40m Algebra 1: Part 1 - Lesson 4**
Multiplication

Materials: Jacobs Algebra

➔ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 3 p.15-16
"Learning the" - "applied to division."

➔ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.17-18

Algebra 1: Part 1

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Term 1

WEEK 2 ⬚ **40m Algebra 1: Part 1 - Lesson 5**
Division

Materials: Jacobs Algebra

→ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 4 p.20-21
"If two people" - "to give x."

→ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.21-22

WEEK 2 ⬚ **40m Algebra 1: Part 1 - Lesson 6**
Raising to a Power

Materials: Jacobs Algebra

→ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 5 p.24-25
"It is an" - "product of n x's."

→ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.25-27

WEEK 3 ⬚ **40m Algebra 1: Part 1 - Lesson 7**
Zero and One

Materials: Jacobs Algebra

→ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 6 p.28-29
"Although the Alexandrian" - "x/1 = x."

→ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.30-31

WEEK 3 ⬚ **40m Algebra 1: Part 1 - Lesson 8**
Several Operations

Materials: Jacobs Algebra

→ READ, NARRATE, & DISCUSS
Ch.1: Fundamental Operations
Lesson 7 p.32-34
"What do you think" - "4 - 4 = 0"

→ PRACTICE
Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.35-36

Algebra 1: Part 1

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Term 1

WEEK 3

⬚ **40m Algebra 1: Part 1 - Lesson 9**

Parentheses

Materials: Jacobs Algebra

→ READ, NARRATE, & DISCUSS

Ch.1: Fundamental Operations
Lesson 8 p.38-40
"Parentheses are" - "10 + 3 = 13"

→ PRACTICE

Complete Exercises and check solutions. Review any concepts necessary.
Set 1 (odd only)
Set 2 (every other question) p.40-42